

ISHAAN MITTAL

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SUMMARY

Software Engineer with deep foundations in Object-Oriented Programming (Java, C++), Data Structures, Design Patterns, and Relational Database Systems (SQL). Experienced across the full software development lifecycle: architecting decoupled microservices, implementing test-driven automation, optimizing RESTful APIs down to 560 ms, and building scalable multi-tiered web systems. Adept at full-stack development, query profiling, and shipping clean, maintainable, production-ready code within Agile sprints.

EDUCATION

Vellore Institute of Technology, Bhopal, India

Sept 2023 – Oct 2027

B.Tech in Computer Science and Engineering (AIML) – CGPA: 8.76/10

SKILLS

- **Core Languages:** Java, C/C++, Python, JavaScript (ES6+), TypeScript, SQL (PostgreSQL, MySQL), Bash Scripting.
- **Software Engineering:** Object-Oriented Design (OOD), Data Structures & Algorithms, Design Patterns, RESTful APIs, TDD.
- **Database & RDBMS:** Relational Schema Design, Query Optimization, Database Indexing, Transactions, Mongoose, Supabase.
- **Web & Frontend:** React.js, HTML5, CSS3, JSON, Asynchronous I/O, Promises, Component Lifecycle, Responsive Design.
- **Tools & Practices:** Git, GitHub, Linux/UNIX CLI, Postman, Docker, Automated Unit Testing, CI/CD, Agile/Scrum Methodologies.

EXPERIENCE

Software Engineering Intern, Phyllo

Oct 2025 – Nov 2025

- Profiled developer integration pipelines across 10+ REST API endpoints, resolving request serialization bottlenecks.
- Instrumented data validation rules and JSON schema checks, ensuring 100% payload reporting integrity across services.
- Authored automated endpoint troubleshooting test harnesses, reducing developer integration issue resolution time by 20%.

Software Engineer Intern (Systems / AI), amasQIS.ai

Mar 2025 – Sept 2025

- Engineered data annotation and telemetry pipelines, optimizing runtime loops to sustain 99.9% uptime across 1,000+ sessions.
- Integrated backend processing scripts into 2 production pipelines, reducing end-to-end operational latency by 40%.
- Automated continuous integration testing suites and regression checks, lifting overall system pipeline reliability by 30%.

PROJECTS

FINOVA – Multi-Tiered Cloud Financial Platform (PostgreSQL, Supabase, Next.js, Java Principles)

Live

Project

- Architected a multi-tiered platform applying OOP principles and Row-Level Security, isolating multi-tenant transaction datasets.
- Optimized RDBMS execution plans via composite indexing and connection pooling, ensuring 99.9% uptime and sub-200ms latency.
- Built automated document data pipelines using Gemini Vision, reducing manual customer transaction entry overhead by 75%.

ShopSense – Event-Driven Voice Commerce Platform (Java/Python, TypeScript, PostgreSQL)

Live Demo

- Engineered an asynchronous service implementing Factory and Strategy design patterns to process real-time streams and intent schemas.
- Built a high-throughput routing layer across 3 decoupled execution modules, sustaining sub-200ms round-trip execution.
- Synchronized client-side state with a PostgreSQL relational database, preserving ACID compliance across concurrent cart updates.

PUBLICATIONS

Ishaan Mittal et al. “Clinical Decision Support through Machine Learning: Benchmarking LightGBM and Deep Learning Models for Diabetes Onset Prediction,” *IEEE ICAAIC 2025*. Benchmarked algorithmic execution complexity across 5-fold cross-validation on clinical datasets; evaluated inference bottlenecks, achieving 3x faster execution and a 0.79 F1 score under resource-constrained system profiles.

LEADERSHIP & CERTIFICATIONS

- **Runner-Up & Most Innovative Idea**, Repdox: Engineered offline-resilient data synchronization architecture for Bharat Pulse AI.
- **Bharatiya Antariksh Hackathon 2026**, ISRO: Built AI graph analytics for road network extraction in a 4-person team.
- **Cybersecurity Analyst Certificate**, IBM (Jun 2025): Mastered 5 formal domains in systems security, access controls, and defense.
- **Vice President**, Music & Band Club (VIT Bhopal): Directed logistics, operations, and multi-team execution for 15+ campus events.